

AlignSpace

Business Proposal

Turn “I like this” into a living-room brief that the homeowner and the interior designer both sign — before the first concept, not after the third revision.



Make "I like this" clear enough to design.

A two-sided agent workspace that turns a homeowner's references and a designer's constraints into one living-room brief both people approve.

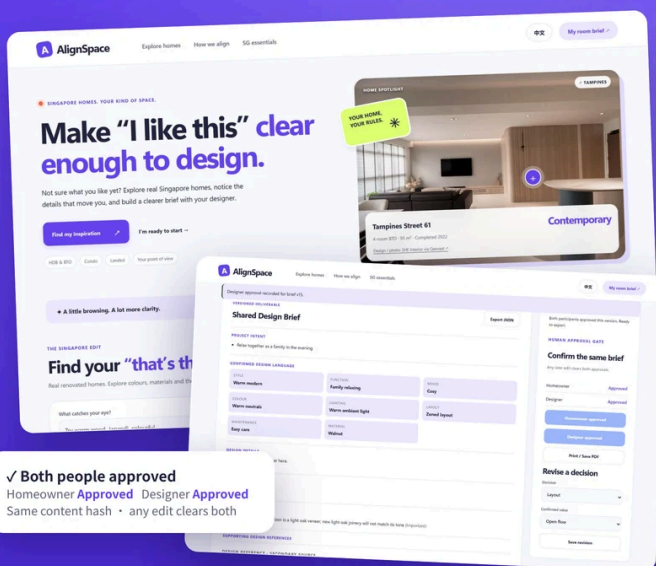
Homeowner

Designer

Adaptive questions

Conflicts stay visible

Dual approval



✓ Both people approved
Homeowner **Approved** Designer **Approved**
Same content hash · any edit clears both

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Team	Four Wolf Kings · team code 8QFDUS2I
Members	Chen Xiaoming · Xing Yiyuan · Xu Tengyang · Xu Zhaobin
Problem statement	Design Inspiration — homeowners find it hard to put design preferences into words and spend time collecting inspiration; designers compile references by hand; misaligned expectations lead to repeated design revisions before both sides agree.
Product	Live at https://54.255.93.19.sslip.io (AWS Lightsail, Singapore region) · code at github.com/JaspinXu/AlignSpace
Companion document	Technical Document — architecture, agents, guardrails, evaluation and AWS deployment evidence
Date	26 September 2026

Evidence labels used throughout: **MEASURED** observed in our build or live deployment · **SOURCE** published figure, cited in the references · **ASSUMPTION** a planning input we chose · **HYPOTHESIS** to be tested in the pilot. We do not claim any time-saving, satisfaction or revenue result yet.

01 Executive summary

Every renovation starts with a conversation about taste. In a small interior-design (ID) firm that conversation is mostly screenshots in a chat, adjectives such as “warm” or “hotel-like”, and a designer decoding them before the first concept. When the two sides read the same picture differently, the gap surfaces later as another revision, a delay or a dispute — usually before a contract is signed and the time can be billed.

AlignSpace is a two-sided agent workspace for that first mile. The homeowner browses real Singapore homes and records notes and must-avoid items; agents turn that evidence into proposals, ask one high-value question at a time, and surface conflicts with the designer’s practical constraints. Nothing counts until a person confirms it, and the brief is released only when both people approve the same version.

787
renovation-contractor complaints to CASE in 2025 — the 4th most-complained industry ^[1] [SOURCE](#)

≈45,600
HDB homes completed (≈19,600) or resold (26,042) in 2025 — each a likely renovation ^{[2][3]} [SOURCE](#)

S\$50–82k
expected 2026 renovation cost of a 4-room HDB flat, new (BTO) to resale ^[4] [SOURCE](#)

US\$0.012
model cost of one note analysis, from tokens measured on our live deployment ^{[5][6]} [MEASURED](#)

Proven now [MEASURED](#)

- Running on the organiser’s AWS Lightsail instance (Singapore) over HTTPS since 21 September.
- 19 of 19 two-session browser checks passed against the live site on 26 September — golden path plus adversarial cases such as prompt injection, invitation replay and role spoofing.
- 77 of 77 automated tests pass; continuous integration on every push.
- A real Claude Sonnet 4.5 analysis on the public site turned a homeowner note into five evidence-linked proposals, although the note contained none of the option words.

To prove in the pilot [HYPOTHESIS](#)

- ≥ 80% of required decisions read the same way by homeowner and designer.
- Designers review a draft brief in ≤ 10 minutes (median).
- ≥ 60% of invited homeowners reach a draft brief.
- Fewer concept revisions than the firm’s own baseline.
- Firms will pay about S\$49 per designer per month.

Our ask. One or two Singapore ID firms with 3–20 designers to run a free, six-week pilot on 10–20 consenting living-room projects, measured against their current chat-and-screenshots process.

02 The problem, as an SME feels it

PROBLEM STATEMENT · DESIGN INSPIRATION

Homeowners find it hard to communicate design preferences in words and spend time collecting inspiration; designers compile references manually. Misaligned expectations lead to repeated design revisions before both sides agree.

How it plays out in a small firm

Stage	What happens today	What it costs
Lead	The homeowner sends 10–30 screenshots from inspiration sites and social media, plus “we like warm and cosy, not too dark”.	No record of <i>which</i> element of each image matters.
Consultation	The designer interprets on the spot and remembers rather than records.	Interpretation lives in one person’s head.
First concept	“I liked the lighting in that photo, not the walnut.”	A revision round, usually unbilled before contract.
Constraints	Existing floors, pets or low-maintenance needs surface after the homeowner has fallen for a look.	Disappointment, renegotiation, lost trust.
Later	“That is not what we agreed.”	No shared record of what was confirmed, by whom, or when.

Why it persists

- **An image carries many attributes.** “I like this” does not say whether the person likes the oak, the light, the layout or the whole room.
- **Words mean different things to each side.** “Warm”, “minimalist” and “hotel-like” are interpreted differently by a first-time homeowner and a trained designer.
- **Constraints arrive late**, and no tool puts a practical constraint next to the preference it contradicts.
- **Nothing records agreement.** Chat threads and mood-board PDFs do not show what was confirmed, rejected or still open.

Why it matters now

- **Renovation remains a top consumer-complaint category.** CASE received 787 complaints about renovation contractors in 2025 (962 in 2024), the fourth-highest industry ^[1]. Not every complaint is about expectations, but a jointly approved record of requirements is exactly the documentation that helps both sides when expectations are questioned.
- **The industry is investing in trust.** CaseTrust had accredited 268 renovation businesses by January 2026, with a target of 500 by 2027 ^[1]. Firms that compete on professionalism and documentation are natural early adopters.
- **The volume is steady and the stakes are high.** HDB completed about 19,600 flats in 2025 and is on track for about 55,000 in 2025–2027 ^[2]; 26,042 resale applications were recorded in 2025 ^[3]. A 4-room renovation is expected to cost S\$50,000–60,000 for a new flat and S\$70,000–81,600 for a resale flat in 2026 ^[4].

03 Customers and users

Role	Who	Job to be done	What they get
Buyer	Owner or design lead of an ID SME with 3–20 client-facing designers	“Help my team align projects without adding operational risk.”	Consistent intake, fewer unbilled revision loops, an auditable approval trail.
User	Designer (“Arjun”)	“Turn scattered references into a usable brief with fewer clarification loops.”	A structured brief with constraints and open decisions visible before the first concept.
User	Homeowner, often a first-time renovator (“Mei”)	“Help me show what I mean and confirm that my designer understands it.”	Plain-language questions in English or Chinese, and control over every suggestion.

Beachhead. Pre-concept alignment for living rooms in HDB and condominium projects, started by the designer after lead qualification. The living room carries rich preferences (style, colour, material, light, layout, mood, function, maintenance) with fewer safety-critical issues than kitchens and bathrooms.

04 The solution

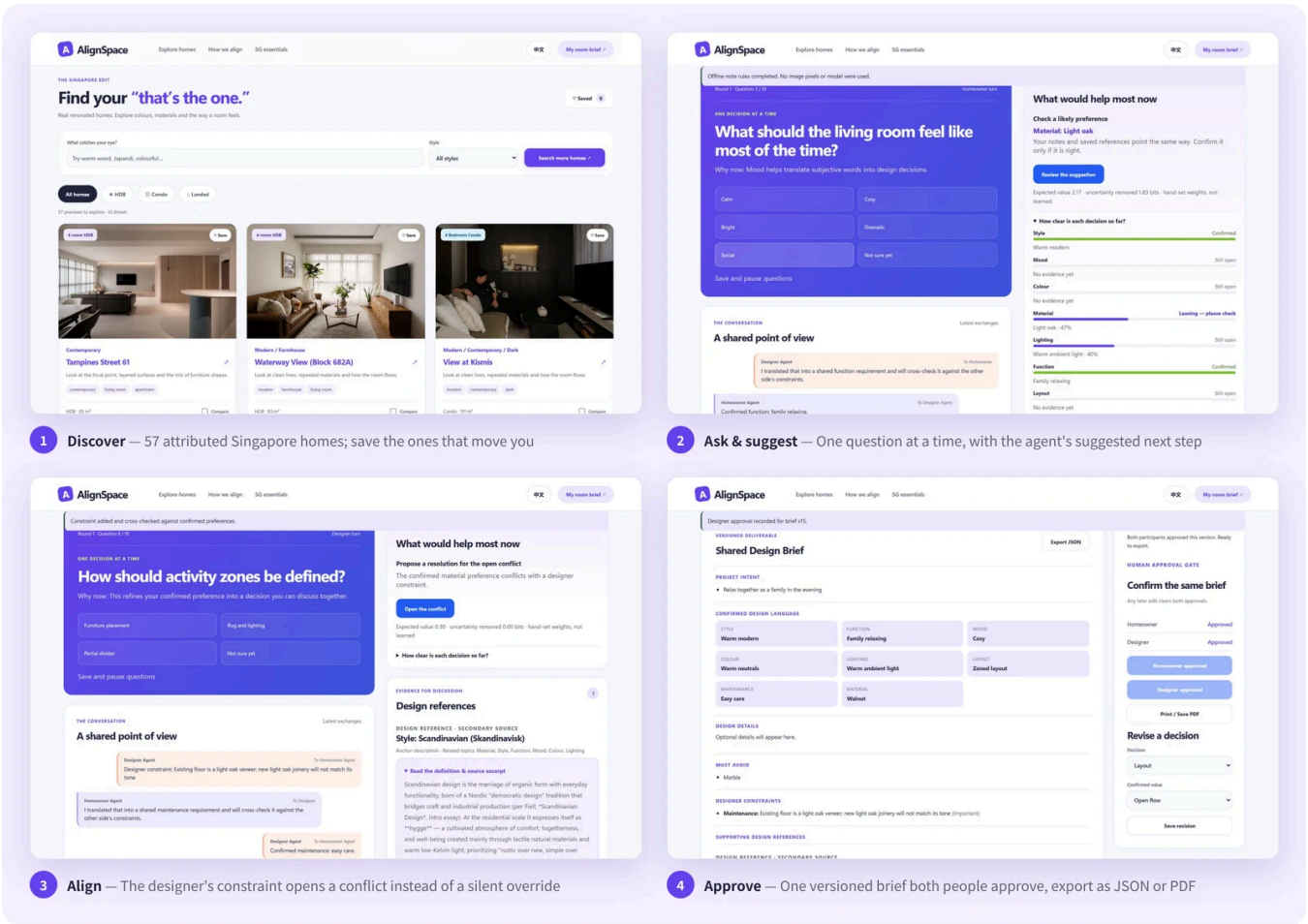


Figure 1. Discover real Singapore homes; answer one question at a time with a suggested next step; see a designer constraint open a conflict instead of a silent override; approve one versioned brief together. Screens from the running product with synthetic data.

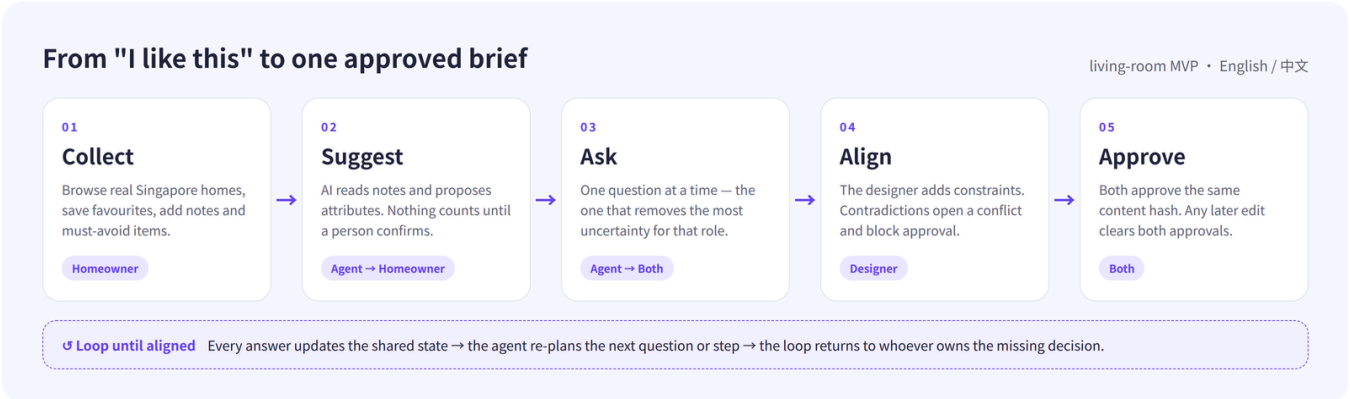


Figure 2. The alignment loop. Agents propose and ask; people confirm, resolve and approve; the loop returns to whoever owns the missing decision.

Pain today	AlignSpace response	Evidence
"I like this" is ambiguous	Notes become attribute-level proposals that quote their source; the homeowner confirms or rejects each one.	MEASURED live call returned 5 proposals, each quoting the note
Words mean different things	One question at a time with plain options and "not sure" always available; the agent picks the question that removes the most uncertainty.	HYPOTHESIS ≥ 80% alignment in the pilot
Constraints arrive late	The designer joins from a private single-use link; a constraint that contradicts a confirmed preference opens a visible conflict and blocks approval.	MEASURED live checks: conflict opened, approval blocked
Nothing records agreement	A versioned brief; both approve the same content fingerprint from their own sessions; any later edit clears both approvals; JSON and PDF export.	MEASURED live checks: same fingerprint, edit clears approvals
AI could overreach	Suggestions stay "proposed"; instructions hidden in notes are ignored; no construction, pricing or compliance advice.	MEASURED live check: injected instruction not executed

What AlignSpace is not: an AI render generator, a quotation or contract tool, or construction, structural, electrical or regulatory advice. It sits upstream of those tools and exports a brief they can use.

05 Market and competition

Demand. About 45,600 HDB households a year start from a newly completed or resale flat (2025: about 19,600 completions ^[2] plus 26,042 resale applications ^[3]), before condominiums and landed homes. **Supply.** CaseTrust-accredited renovation businesses number 268 today and are targeted to reach 500 by 2027 ^[1], alongside many non-accredited firms. We size the opportunity by *firms*, because firms pay.

Illustrative scenario	ASSUMPTION	Paying firms	Designer seats (5 per firm)	Price per seat / month	Annual recurring revenue
Year 1 — design partners convert		10	50	S\$49	S\$29,400
Year 2 — early adopters		40	200	S\$49	S\$117,600
Year 3 — referral-led growth		100	500	S\$49	S\$294,000

Arithmetic on our assumptions, not a forecast. For scale, the Year-3 scenario equals one-fifth of the CaseTrust target of 500 accredited renovation businesses ^[1]. Expansion beyond the beachhead: other rooms (with added professional review for wet areas), whole-home briefs, and other markets where homeowners and designers work across English and Chinese.

Competitive landscape

Alternative	Good at	Leaves open	AlignSpace position
Chat, screenshots and PDF mood boards (status quo)	Fast, familiar, free	No structure; no record of what was confirmed; constraints arrive late	Replaces the ad-hoc intake; chat stays for conversation
Inspiration platforms such as Pinterest, Houzz and Qanvast	Discovery and a huge supply of images	A saved image is not a stated preference; one-sided	Links out to attributed homes and treats saves as weak evidence only
AI image and render generators	Inspiring visuals quickly	Pictures, not requirements; can set expectations the site or budget cannot meet	Produces agreement on requirements; visuals remain the designer's craft
Quotation, CRM and project tools for renovation firms	Downstream execution	Assume the brief already exists and is agreed	Upstream of them; exports schema-valid JSON

Defensibility. Not the base model — any capable model can sit behind our adapter. What compounds is the two-party preference and conflict data (with consent), the question policy and its evaluation set, bilingual design vocabulary, and the approval step embedded in a firm’s intake.

06 Business model and unit economics

Pricing **HYPOTHESIS**

Plan	Price	For
Pilot	Free, 6 weeks	1–2 design partners
Studio	\$49 per designer seat per month	Firms; unlimited homeowners, fair-use analysis cap per project
Per brief	\$15 per dual-approved brief	Freelancers and low volume

Break-even for a firm. At \$49 per seat, the tool pays for itself if it saves a designer about one hour a month **ASSUMPTION** \$50 loaded hourly cost — about ten minutes per project at six projects a month. The pilot measures review time and revision counts against the firm’s own baseline.

Cost to serve

Item	Cost
One note analysis: 1,873 input + 426 output tokens MEASURED × Claude Sonnet 4.5 list price US\$3 / US\$15 per million ^[6]	US\$0.012
Model cost per project at 1–5 analyses ASSUMPTION ; hard cap of 20 per project	US\$0.01–0.06 cap ≤ US\$0.24
Hosting: one Lightsail instance, 2 vCPU, 4 GB, 80 GB ^[7]	US\$24 / month
Per seat per month: 6 projects × 3 analyses, 50 seats sharing one instance ASSUMPTION	≈ US\$0.70

On 26 September the live instance used 0.9 of 3.8 GB memory and 6.8 of 77 GB disk **MEASURED** . Infrastructure is small relative to price; the real costs are onboarding, support and sales, which the pilot will size.

07 Go-to-market

PHASE 0 · OCT–DEC 2026

Design partners

One or two ID SMEs through our network and hackathon-ecosystem introductions. Free six-week pilot, 45-minute onboarding, weekly metric and safety review. No CRM integration required.

PHASE 1 · H1 2027

Early adopters

Quality-focused and CaseTrust-accredited studios, reached through pilot case studies, designer referrals and bilingual “how to brief your designer” guides for homeowners.

PHASE 2 · H2 2027 ONWARD

Scale

JSON export into quotation and CRM tools, partnerships with inspiration platforms, more rooms, and a firm dashboard that tracks alignment across projects.

How a firm adopts it. The designer sends the project link after lead qualification → the homeowner completes the guided brief at home → the designer reviews the summary before the first workshop → the pair resolves open decisions and approves → the brief is exported into the firm’s existing workflow. The homeowner never needs an account, and the firm keeps the relationship.

08 Pilot plan and measurable outcomes

Set-up. Six weeks; one or two SMEs; 10–20 consenting living-room projects; baseline from the firm’s recent comparable projects run the usual way. Synthetic or consented data only; no purchasing, contact or contractual output.

Metric	HYPOTHESIS	Definition	Target	How we measure
Brief Alignment Rate		Share of 20 rubric decisions where homeowner and designer independently pick the same interpretation after the session	≥ 80%	Blind rubric at session end
Invitation-to-draft		Invited homeowners who reach a draft brief	≥ 60%	Audit log
Designer review time		Median minutes to review a draft brief	≤ 10 min	Timestamps plus designer log
Actionability		Designers rating the brief at least 4 of 5 as a starting point	≥ 70%	One-question survey
Revisions		Concept revisions before sign-off, compared with baseline	Fewer	Designer-reported per project
Willingness to pay		Designers who would pay S\$49 per seat per month	≥ 50%	Exit interview
Safety		P0 privacy or security incidents	0	Incident log

Decision rules. Convert and expand if at least four targets are met with no P0 incident; iterate on the weakest step if two or three are met; stop or pivot if one or none is met, or if a P0 incident is not resolved.

09 Feasibility and scalability

The product already runs end to end on AWS with bounded AI usage, security controls and English/Chinese copy (details and evidence in the Technical Document). The path from pilot to scale changes infrastructure, not the product's logic:

Concern	Today (pilot)	At scale
Data store	SQLite in WAL mode on one instance; backup before each release switch	Amazon RDS for PostgreSQL with per-firm tenancy and scheduled backups
Images	Private disk storage; images decoded, resized and metadata-stripped	Amazon S3 with signed URLs and lifecycle deletion
Compute	One Lightsail instance (US\$24 per month) behind Caddy with HTTPS	Two or more instances or containers behind a load balancer
Model	Organiser Claude Sonnet 4.5 JSON API with per-project and daily caps	Amazon Bedrock under the firm's account; a regional endpoint if data residency requires it (+10% over global ^[6])
Identity	Browser-session access with single-use designer invitations	Firm accounts and single sign-on, recovery and admin roles
Vision	Adapter built but disabled: the gateway failed a known-image test	Enabled once a vision endpoint passes the same test
Compliance	Consent before upload, data minimisation, deletion by design	PDPA review, data-protection officer, data-processing agreements with firms

10 Risks and mitigations

Risk	Mitigation
Designers see it as replacing their relationship work	It is the designer's intake tool: the designer sends the link, owns the constraints and exports the brief into their own workflow.
Homeowners will not do pre-work	At most ten questions per round, pause and resume, "not sure" always allowed, a guided sample; completion is a pilot metric.
The AI misreads a preference	Proposals never auto-confirm, evidence is quoted, rejection is one click, and an explicit offline mode exists.
Liability for construction or safety claims	Out of scope by design; critical constraints require professional review before approval, and the product says so.
Privacy (PDPA)	Consent before upload, private storage, minimisation and deletion; no training on project data without separate consent.
Model cost or provider dependency	A provider adapter (Anthropic, OpenAI and Ollama formats), per-project and daily caps, cooldowns and an offline mode.
Large platforms add similar features	Focus on two-sided agreement data and workflow embedding; complement inspiration platforms rather than compete with them.

11 Social impact

- **Taste without jargon.** People without design vocabulary express preferences by reacting to real homes and answering plain questions, in English or Chinese, including mixed-language notes.
- **Transparency.** Every suggestion shows its evidence, and open decisions stay visible instead of being guessed.
- **Fewer misunderstandings.** A jointly approved, versioned record of requirements supports better conversations when expectations are questioned.
- **Inclusion.** Works in any browser with no app install; a manual path is always available; accessibility targets include WCAG 2.1 AA contrast and keyboard use.

12 Roadmap

When	Milestones
Now · Sep 2026	Living-room product live on AWS; two-sided approval; English and Chinese; bounded Claude Sonnet 4.5 note analysis; 77 tests and 19 live checks.
Next · Q4 2026	Design-partner pilot; paired homeowner–designer trials; calibrate the belief model on pilot data; enable image understanding once a provider passes the known-image test; account recovery and HSTS.
Later · 2027	Whole-home briefs with professional review for wet areas; firm dashboard; integrations with quotation and CRM tools; Malay and Tamil copy.

13 Team and honest status

Four Wolf Kings (8QFDUS2I) — Chen Xiaoming, Xing Yiyuan, Xu Tengyang and Xu Zhaobin — built AlignSpace for the NUS-ISS Show Me Your Agents hackathon (public category).

What we claim	Evidence
A working two-sided product on AWS	MEASURED Live URL; deployment snapshot and 19/19 live checks on 26 September (Technical Document §9)
Guardrails that hold under adversarial checks	MEASURED 19 two-session checks and 77 automated tests
Real model integration with bounded cost	MEASURED Live call record with model, prompt version, tokens and latency (21 September)
Business outcomes	HYPOTHESIS Not yet measured. No user study, time saving or revenue is claimed; the pilot above is designed to measure them.

— References

1. Consumers Association of Singapore (CASE), media release, 9 February 2026: renovation contractors 787 complaints in 2025 (962 in 2024), fourth-highest industry; 268 CaseTrust-accredited renovation businesses as of January 2026, target 500 by 2027. case.org.sg
2. Mothership, 8 January 2026, reporting HDB: about 19,600 flats completed in 2025; about 55,000 flats from 2025 to 2027. mothership.sg/2026/01/hdb-bto-2026
3. ERA Singapore, 2 January 2026, HDB flash estimates: 26,042 resale applications in 2025 (28,986 in 2024). era.com.sg
4. Qanvast, 12 February 2026, expected 2026 renovation costs from 2025 medians across 341 BTO and 449 resale projects: 4-room BTO S\$50,000–60,000; 4-room resale S\$70,000–81,600. qanvast.com
5. AlignSpace live model-call record, 21 September 2026: 1,873 input and 426 output tokens, one attempt, 7.5 s. [docs/evidence/gateway-live-check.json](https://docs.evidence/gateway-live-check.json)
6. Anthropic, Claude pricing: Claude Sonnet 4.5 US\$3 per million input and US\$15 per million output tokens; regional endpoints 10% above global. platform.claude.com

7. Amazon Lightsail pricing: Linux instance with 4 GB memory, 2 vCPUs, 80 GB SSD and 4 TB transfer, US\$24 per month.

aws.amazon.com/lightsail/pricing

All web sources accessed 26 September 2026. Figures from sources are reported as published; scenario and cost-per-seat figures are our arithmetic on the assumptions stated beside them.